Student's Name:

……

P525/2

CHEMISTRY

(Practical)

Paper 2

$3^1/_4$ Hours

Resourceful mock

Uganda Advanced Certificate of Education

CHEMISTRY

Paper 2

Practical

3 HOURS 15 MINUTES

INSTRUCTIONS:

- The paper consists of TWO compulsory items. Answers to these items are to be written in the spaces provided in this booklet. Use **blue** or **black** ink
- All working must be clearly shown.
- Mathematical tables and silent non programmable scientific calculators may be used.
- Reference books are not allowed such as textbooks and summaries.
- Candidates are advised to carefully read each item and ensure that all the apparatus and chemicals required for the experiment are available so that they can plan appropriately.

Where necessary use; (C=12, H=1, O=16,Na=23, density of solution= 1.0gcm^{-3}, specific heat capacity of solution= 4.2jg^{-10}c^{-1})

Examiner's use only.

Codes									
Item 1									
Item 2									

ITEM ONE:

A metal fabrication company in Wakiso District uses hydrochloric acid to remove rust from magnesium alloy parts before coating.

The quality control manager is concerned because the concentration of the hydrochloric acid must be known accurately to control the rate of reaction and gas production. Also, the company wants to know the heat energy released when magnesium reacts with the standardized acid, because excess heat above **60,000J** can damage equipment and affect worker safety.

To ensure safety and efficiency, the laboratory technician has been asked to verify the concentration of the acid first. However, the acid label was damaged and its concentration is missing. The technician has great conviction that the acid's concentration is between **0.95-1.05M**. The company has a fresh bottle of **Solid T**; ***anhydrous sodium carbonate***, **Na_2CO_3**, which is a primary standard and can be used to standardize the acid.

After standardizing the acid, the technician will then use it to react with a known mass of magnesium powder where the temperature change will be used to determine the molar heat of displacement of the reaction between magnesium alloy parts and hydrochloric acid.

The head of the science club and his chemistry learners at a nearby secondary school has been invited to observe and assist with the analysis.

You have been provided with:

FA1; Hydrochloric acid of unknown concentration.

Solid Z; Magnesium alloy parts (weigh exactly 1.2g)

Solid T; Anhydrous sodium carbonate, ***Na_2CO_3*** *(weigh exactly 5.8g of T and use it to make a 250cm^3 solution. Label it* ***FA2****)*

Task:

As a learner of chemistry;

Plan, design and carry out an experiment to help the company and use your results to advise the company management accordingly.

(Your report should include an Aim, Hypothesis, variables, Procedure, Risks and Mitigations, Data presentation, Analysis, Conclusion and Recommendation)

ITEM 2:

Roofing sheets from an old factory in Jinja industrial park showed extensive corrosion after years of exposure to acid rain from industrial fumes. A white solid corrosion product was collected by environmental scientists who believe that corrosion involved reaction of the metal roofing material and the acidic pollutants from the atmosphere. Critical Laboratory analysis shows that the solid is likely to contain two different metal ions and two different anions from a mixture of soluble and insoluble salts. The environmental scientists need to identify the actual ions present to be able to treat this product correctly to avoid environmental pollution.

Your chemistry teacher has asked you to help out in this investigation

You are provided with the following;

Solid Y which is a sample of the solid corrosion product

Relevant apparatus and Laboratory reagents.

Task:

As a chemistry learner;

Design an experiment to determine the cations and anions present in the **solid Y** and advise the environmental scientists accordingly.

(Your report should include; Procedure, Data presentation and interpretation, Analysis, Conclusion and Recommendation).

END

CONFIDENTIAL

On additional to the common reagents and apparatus used in chemistry laboratory, each student should be provided with the following;

1 burette of 50cm^3 capacity

1 pipette of 25cm^3 (or 20cm^3)

2 filter papers

1 thermometer

2 conical flasks

1 funnel

1 volumetric flask of 250cm^3 capacity

6 test tubes

3 plastic beakers

100cm^3 of FA1

6.5g of T

2.5g of Y

2.0g of Z

Easy access to;

Weighing scale

Heat source

Indicators

Reagents to identify cations, anions, and gas(es) evolved

- **FA1** is made by dissolving **83.5cm^3** of 37% or 36% hydrochloric acid; (**HCl**) in 1 litre of distilled water to make **1.0M HCl** solution.
- **Solid T** anhydrous sodium carbonate
- Solid **Z** is magnesium powder;
- Substance **Y** is a solid mixture of **Zinc sulphate : Barium carbonate OR Calcium carbonate** in the ratio **2:1**